

Project Name:

Online Banking Web Application

1. Introduction

The Online Banking Web Application allows customers to:

- Login securely
- View account balances
- Transfer funds
- View transaction history
- Logout securely

The system must support secure authentication and transaction validation.

2. Functional Requirements

2.1 User Login

- User shall login using:
 - Username
 - Password
- Username must be unique.
- Password must contain:
 - Minimum 8 characters

- One uppercase letter
 - One number
 - One special character
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- After 3 failed login attempts:
 - Account should be locked for 15 minutes.
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- System shall display proper error messages for:
 - Invalid username
 - Invalid password
 - Locked account

2.2 Dashboard

- After successful login:
 - User shall see account summary.
 - Available balance should be displayed.
- Dashboard should load within 3 seconds.

2.3 Fund Transfer

- User shall transfer money between accounts.

Required fields:

- Beneficiary account number
- IFSC code
- Transfer amount

Validation rules:

- Amount must be greater than 0.
- Amount cannot exceed available balance.
- IFSC code must be valid.
- Beneficiary account must exist.

2.4 Transaction Confirmation

- User shall receive confirmation message after successful transfer.
- Transaction ID must be generated uniquely.

2.5 Logout

- User shall logout securely.
- Session should expire automatically after 10 minutes of inactivity.

3. Non-Functional Requirements

- System shall support 10,000 concurrent users.
 - Sensitive data must be encrypted.
 - Application should support Chrome, Firefox, and Edge browsers.
 - API response time should be under 2 seconds.
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4. Security Requirements

- SQL Injection protection required.
 - XSS protection required.
 - Passwords must be hashed.
 - OTP verification required for large transactions above \$10,000.
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5. Error Handling

- System shall display user-friendly error messages.
- Logs should be generated for failed transactions.